TorqSiege — Collapse Test Protocol v1

2026-09-04. The fastest path to a yes/no on TorqSiege's one gating unknown: ***will a speed-limited light RC buggy topple a structure dramatically enough on camera to carry an event?*** *Everything in [[TorqSiege — Build Phase & Weapons v1]] is designed around this. Don't build the event until this passes. This is a kitchen-table test, not a production — designed to be done in* ***one afternoon, near-zero cost, with one car.***

The question we're answering

Not "can a car knock a block over" (yes, trivially). The real bar: **does the collapse look bigger than the hit** — a wall that *cascades*, blocks flying, a slow-mo money-shot — at the *low* mass/speed our safe rental cars run at? If only a heavy ram at unsafe speed can do it, TorqSiege needs a rethink. This test finds the **cheapest structure recipe that collapses dramatically at the lowest car energy.**

Fastest-possible principle

- **Don't wait for the fleet or for 3D prints.** Use any RC car you can get *today* (a borrowed/owned hobby buggy, or even a cheap toy-grade to start) + recycled cardboard + a phone. Prove the *physics* first; buy hardware only once you know the recipe works.
- **Simulate the heavy ram with stick-on weight** (tape coins/washers/a bolt pack to a light car) instead of buying a 1/10 up front — this brackets "how much mass do we actually need" before spending.
- **DIY the weapons before printing them** — a cardboard/plywood wedge and a bolt-tip ram test the *geometry* in minutes; print the winners later.

What you need (the whole kit)

Have-now / cheap:

- **1 RC car** — ideally the intended workhorse (WLtoys 144001 / MJX 16208), but *any* RC buggy proves the physics. Bonus: a second heavier car OR weights to add.
- **A phone with slow-mo** (240fps if it has it) + a way to mount it overhead (a chair/ladder/box) and one side-on.
- **Structure materials (recycled first):** cardboard boxes/sheets, plastic cups, EPS/thermocol foam blocks, wooden blocks/Jenga, empty tins, foam board.
- **Build tools:** craft knife + steel ruler + cutting mat (for pre-scoring), masking tape, a marker, tape measure, scissors.
- **Floor:** a flat ~3×3 m space (garage/hall/office floor).

Optional (round 2, only if round 1 is promising):

- 3D-printed wedge + ram-spike + grapple (the [[TorqSiege — Build Phase & Weapons v1]] first three) on a universal mount.
- The actual 1/10 siege ram (WLtoys 104019) to confirm the "heavy" end.

The test matrix — structures × weapons × energy

Run every promising **structure recipe** against each **weapon** at each **energy level**, film each hit, score the collapse.

Structure recipes (the real variable — engineer the wall to fall):

1. **Loose-stack blocks** — light blocks stacked, no fixing (foam / cardboard cubes).
2. **Pre-scored cardboard wall** — knife-scored fold lines so it buckles on impact.
3. **Weak-tab modular** — panels held by small cardboard tabs / light tape that release.
4. **Domino logic** — pieces arranged so one topple chains the rest.
5. **Cup/tin pyramid** — classic, instant-satisfying, near-zero cost (baseline "easy" control).
6. **"Too stable" control** — a glued/heavy wall we *expect* to resist, to mark the failure boundary.

Weapons: bare bumper · DIY wedge · DIY ram-spike · (later) printed versions · (later) grapple-pull.

Energy levels: car at low/capped speed · at higher speed · with added weight (sim-ram) · actual 1/10 ram.

Method (per hit)

1. Build the structure to a fixed spec (note height, block count, material).
2. Cameras rolling — **overhead + side, slow-mo on.**
3. One clean run at the target energy.
4. **Score the collapse 1–5:** 1 = barely nudged · 3 = topples but dull · **5 = dramatic cascade, blocks fly, reel-worthy.**
5. Log: recipe, weapon, energy, score, + did the *car* take damage (the other half — see [[Car Fleet, Scale & Protection]] §4).
6. Reset, repeat. ~20–30 hits covers the matrix in an afternoon.

Pass / fail (the decision)

- **PASS** = at least one structure recipe scores **4–5 at safe, capped car energy** with a DIY/cheap weapon, and rebuilds in **<60 s**, and doesn't wreck the car. → TorqSiege is real; lock that recipe, print the winning weapons, grade it in [[Game Ideas - Graded]], build the event.
- **CONDITIONAL** = only the heavy sim-ram / 1/10 hits 4–5. → TorqSiege works but *requires* the 1/10 ram as the attacker; factor its cost + a safety plan.
- **FAIL** = nothing hits 4–5 without unsafe speed. → don't build TorqSiege as-is; pivot (much lighter/taller "jenga-tower" structures, magnetic-release joints, or a partly rigged "assisted collapse" showpiece).

Why this also de-risks TorqFall (the R&D graft)

The **Collapse Gate** round ([[TorqFall — Level Directory v1]] Tier-3 #6) is a small breakable finish wall. Building the winning wall-recipe from this test into that round lets us **field-test the collapse mechanic in front of a live TorqFall crowd** before committing a rupee to a full TorqSiege build — the cheapest possible validation.

Do-it checklist: (1) grab a car + a box of recycled cardboard/cups/foam today; (2) build recipes 1–6; (3) film the matrix, score it; (4) if PASS, print the winning weapons + lock the wall recipe into the castle-stock lines of [[TorqSiege — What The Event Needs]] §1 (the operative BOM — not Build & Fabrication Plan v1); (5) report the winning recipe back here.